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The evaluation of bankruptcy prediction models based on socio-economic costs

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ABSTRACT

Corporate bankruptcies often have severe consequences on all stakeholders, from financial stakeholders losing their investment to employees losing their jobs. Yet traditional bankruptcy prediction models typically focus solely on predicting the event of bankruptcy itself, and do not consider the socio-economic consequences of their prediction. Therefore, this study aims to integrate these perspectives into the machine-learning (ML) modeling process to consider different costs caused by bankruptcy. We improve upon existing bankruptcy prediction models by actively taking the social and financial impacts of bankruptcy into account. Specifically, we consider two alternative evaluation metrics: the financial costs of bankruptcy, and the social impact measured using number of lost jobs as proxy. We compare a variety of machine-learning models as well as multivariate discriminant analysis and logistic regression, the latter serving as a benchmark to show the improvements that can be achieved using ML models. We apply the models on a large real-world data set from the Compustat database, containing listed companies in North America for the period from 1985 to 2020, with over 190,000 company-year observations. Our results show that small differences in statistical performance can translate into large differences regarding socio-economic costs, and that the selection of the 'best' performing model crucially depends on the evaluation metric considered.

1. Introduction

Bankruptcy has multiple negative effects on the investors, creditors, employees, customers, and other stakeholders of the affected company. Previous research suggests that suppliers to economically distressed firms also experience losses (Kolay et al., 2016) and that their strategic partners experience negative stock price reactions (Boone & Ivanov, 2012). "As when a pebble is thrown into a lake and the shockwaves reach far beyond the point of initial impact, when a company becomes financially distressed/insolvent there are adverse consequences for its diverse stakeholder groups…" (Jackson & Wood, 2013, p. 182). As bankruptcy in most cases is not a sudden event, but rather a gradual process, signs of financial distress can be observed years before the event (Ooghe & De Prijcker, 2008). Therefore, if financial distress is indicative of the likelihood of future bankruptcy and can be detected in time, companies should be able to take the required actions to improve their financial health and minimize the negative socio-economic effects on the stakeholders.

Over the past decades, several researchers have suggested statistical and analytical approaches to predict bankruptcies, ranging from simple statistical prediction models (Altman, 1968; Beaver, 1966) to modern machine-learning (ML) models (Geng et al., 2015; Lee & Choi, 2013). However, most of these models focus on the binary event whether a company goes bankrupt or not, without taking additional characteristics of the companies such as debt volume, firm size, or number of employees into account. Yet, in such a binary prediction scenario, these additional company characteristics can be crucial to quantify the benefit, or costs, of the prediction models. For instance, lending to a company that will go bankrupt is usually more costly than not lending to a solvent company that is falsely predicted to be in financial distress or go bankrupt. An investor that lends to a company that goes bankrupt can lose up to 100% of their investment, while not lending to a company which remains solvent only causes the opportunity costs of the lost interest. In other words, the financial implications of Type I and Type II errors are different, and ML models for bankruptcy prediction need to take this into account. While some of the previously suggested models consider these different costs, there is no systematic comparison of the financial costs of different prediction model approaches. Without further specification, binary prediction models also do not distinguish between the bankruptcy of a small company and that of a large company. Specifically, the latter can lead to significantly higher downstream effects such as lost jobs. Furthermore, it is also unclear how differences in model performance measured through traditional

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ML metrics, such as accuracy or area under the ROC curve (AUC), translate into estimated financial or socio-economic costs, for example measured as financial costs of misclassification, or even the number of employees affected.

To address this research gap, we analyze if ML models for bankruptcy prediction can reduce the costs incurred by the investor, compared to a naïve model where every company in the dataset receives a loan. As the companies differ regarding size, assets, debt, and the number of employees, the costs of each misclassification error depend on the individual characteristics of each company. To measure the financial impact of misclassifications, we take these individual company characteristics into account to distinguish between the costs of Type I and Type II errors and calculate them based on Altman et al. (1977). The number of lost jobs that is caused by the unpredicted bankruptcies is a proxy for the social costs, as increased unemployment rates do not only cause financial costs for the individual person, but also lead to increased government spending, increased poverty rates, and reduced purchasing power.

Using a large real-world dataset from Compustat North America on company bankruptcies in the United States, our results show that ML models can lead to a significant cost reduction compared to traditional statistical prediction methods. Specifically, our evaluation shows that ML models for bankruptcy prediction are successful in lowering the investor's incurred costs of bankruptcy by generating more accurate predictions of financial costs. While the statistical models can even lead to an increase in costs compared to a naïve investment strategy (where every company in the dataset receives a loan), the ML models reduce the costs by 40% to 63%, depending on the model and input variables. Further, our findings indicate that models that are similar regarding the standard performance metrics can lead to a considerable difference in cost reduction. Specifically, we see that models with the best performance on standard metrics such as the area under the curve or balanced accuracy do not necessarily correspond to the best models in terms of achieved cost reduction. These results highlight that such a specific evaluation metric, which takes the financial impact of the bankruptcy prediction scenario into account, can yield crucial information in the selection of good prediction models.

We make two main contributions:

- First, we extend existing research on bankruptcy prediction by taking the financial, and indirectly the socio-economic, impacts of bankruptcy into account. Specifically, we introduce two new evaluation measures in addition to the standard prediction metrics for binary classification problems used in previous bankruptcy prediction models: cost reduction and unpredicted job loss.
- Second, we provide a systematic comparison of the performance of different ML approaches on various evaluation metrics. In particular, we consider if small differences in accuracy and other evaluation metrics also translate into comparable performance differences on financial costs and the number of affected employees or not.

We structure our paper as follows. First, the next section provides a brief overview of related work in bankruptcy prediction and ML in general. The following section then provides details about our evaluation methodology, data, models, and evaluation metrics. Afterwards, we provide initial findings using Exploratory Data Analysis (EDA), followed by the evaluation of the prediction models and their performance on different evaluation metrics. We conclude with a discussion of the results and their implications.

2. Related work

2.1. Previous bankruptcy prediction approaches

Corporate bankruptcy is a multidisciplinary problem. Many academic papers from the accounting, finance, management, and information systems fields deal with the reasons, as well as consequences, prediction, and prevention of bankruptcy. Over 25 different approaches to bankruptcy prediction (theoretical, statistical, and machine-learning) have been published so far. Fitzpatrick conducted the first bankruptcy study in 1932 (Fitzpatrick, 1932). He conducted a matched sample of 19 bankrupt and 19 non-bankrupt companies. He did not perform any statistical analysis, but compared different financial ratios between those firms. The first bankruptcy prediction model was published by Altman (1968), using a sample of 33 bankrupt and 33 non-bankrupt companies matched by size and industry. Based on five financial ratios, he performed a multivariate discriminant analysis and calculated a bankruptcy probability score. The corresponding prediction model classified 63 of the 66 companies correctly. Even 50 years after its publication, it still serves as a benchmark for the performance of different bankruptcy prediction models and the five ratios are still used as a starting point for new models. Ohlson (1980) provided another important model. He used a much larger sample and performed a linear regression analysis with nine different financial ratios. Zmijewski (1984) introduced the first probit model and tested different potential biases caused by sample selection or data collection procedures. The results suggest that biases exist, but do not have an impact on the statistical inference or classification rates. All those models achieve a relatively high accuracy, while using a relatively small sample and a limited number of financial ratios as explanatory variables.

The academic interest in bankruptcy prediction particularly increased after the financial crisis in 2007. With the increased popularity of machine-learning techniques in recent years, they became the standard for bankruptcy prediction, as statistical models such as logistic regressions and discriminant analysis have some disadvantages - for instance, they often rely on strict assumptions. With machine-learning, it became possible to include new variables in the models which could not be included in the analysis before (for instance, corporate governance indicators, textual data from the annual report, newspaper articles about the company, etc.). Machine-learning models are suitable for financial data as they do not rely on assumptions about the data distribution and allow for nonlinear relationships, which are likely to exist between the probability of bankruptcy and different financial ratios (Giordani et al., 2014). Six out of the ten most cited papers about bankruptcy prediction use some machine-learning technique (Shi & Li, 2019), and neural network-based approaches seem particularly popular. The predictions from ML models are often more accurate than those from regression or discriminant analysis approaches, and they can typically achieve an accuracy greater than 90% (Alaka et al., 2018).

In general, most of the models use accounting or financial market data. "The use of ratios is as much due to their predictive power as to their availability and standardization. They generally allow for good discrimination between failed and non-failed firms (Altman, 1968), are easily available and are homogeneous because they are calculated in the same way within a given regulatory framework [...] Accounting models thus dominate the world of bankruptcy prediction" (du Jardin, 2016, p. 237). Alfaro et al. (2008) find that boosted decision trees achieve a lower error rate than artificial neural networks when the industry code and accounting ratios are used as input variables. Chen et al. (2010) use a novel KMV-based model to analyze the credit risk in Chinese listed SMEs. They find that the credit risk in China is high with an increasing tendency, and that the asset size has the largest impact on the credit risk. Zelenkov and Volodarskiy (2021) argue that it is optimal for bankruptcy prediction models to have a balanced ratio of false positive and false negative predictions, as those two errors have similar consequences. They propose a Multi-Objective Classifier Selection algorithm which selects only classifiers that belong to the Pareto optimal set regarding the ratio of false positive and false negative predictions. They test different methods of undersampling, oversampling and no sampling methods and use a Russian and a Polish dataset. Their approach showed a significant improvement in terms of different metrics (geometric mean, area under the curve). Olson et al. (2012) analyze the tradeoff between the accuracy and transparency

Table 1
Overview of bankruptcy prediction models with variables and evaluation metrics considered. AUC refers to the area under the ROC curve.

Model	Type of prediction model	Variables	Evaluation metrics
Altman (1968)	Discriminant analysis	Accounting	Accuracy
Ohlson (1980)	Logistic regression	Accounting	Accuracy
Zmijewski (1984)	Probit	Accounting	Accuracy
Hillegeist et al. (2004)	Black–Scholes–Merton, Option-pricing model	Market	Log-likelihood, Pseudo-R^2
Alfaro et al. (2008)	Neural networks, Boosted decision trees	Accounting, Industry codes	Error rate
Chen et al. (2010)	KVM	Accounting, Market	AUC, Mean-difference
Olson et al. (2012)	Decision trees, Neural networks, Logistic regression, Support vector machines	Accounting, Market	Accuracy
Jackson and Wood (2013)	Discriminant analysis, Options, Neural networks, Logistic regression	Accounting, Market, Macroeconomic	AUC, Accuracy
Barboza et al. (2017)	Machine-learning models	Accounting, Market	AUC, Accuracy
Tobback et al. (2017)	Smoothed wvRN, Support vector machines	Accounting, Relational	AUC
Zelenkov and Volodarskiy (2021)	Logistic regression, Machine-learning models	Accounting, Macroeconomic	Geometric mean, AUC

of bankruptcy prediction models. They find that decision trees are relatively more accurate than neural networks, but have more rule nodes than desired. Tobback et al. (2017) find that a combination of financial and relational data improves the performance of bankruptcy prediction models, as a company is more likely to fail when one of its related companies already filed for bankruptcy. Barboza et al. (2017) compare the performance of different bankruptcy prediction models based on the sensitivity, specificity and area under the curve, and find that machine-learning models outperform logistic regressions and linear discriminant analyses, as well as that random forests show the best performance among the machine-learning models. Our analysis builds on their approach and extends their work by introducing different costs of Type I and Type II errors.

Table 1 shows that, although different bankruptcy prediction approaches in the past used a variety of methods and variables, the most common evaluation metrics are statistical, such as accuracy, geometric mean, and area under the ROC curve (AUC).

2.2. Cost-sensitive evaluation metrics

Bankruptcy prediction models are often evaluated based on standard evaluation metrics for (binary) classification problems, without taking different costs of misclassifications or other financial effects into account. Yet this financial, or cost-sensitive, perspective has been considered for prediction models in other business domains and applications before, e.g. for the evaluation of stock-price prediction models based on the profit (Al Nasseri et al., 2015; Chen & Chen, 2016; Krauss et al., 2017). Al Nasseri et al. (2015) compare the return of different stock-trading strategies based on the sentiment of the posts in a stock micro-blogging forum. Chen and Chen (2016) analyze the profitability of a pattern-recognition model that forecasts the turning points of stock prices. Krauss et al. (2017) compare the performance of different ML models that predict the probability that a stock outperforms the general market based on the daily return rate.

Meanwhile, bankruptcy prediction models usually assume that the costs of all errors are equal. This makes it impossible to assess the real costs, or profit, those models could generate. Agarwal and Taffler (2008) compare different models for credit-risk prediction based on different approaches, one of them being the economic value of the model for the bank. However, they assume that all the loans are the same size and the loss given default (LGD) is the same for all companies. The same approach can be found in Blöchlinger and Leippold (2006). They treat the loss given default as an exogenously determined constant value. Altman et al. (2020) compare different ratios of Type I and Type II errors of bankruptcy prediction models, while Neves and Vieira (2006) state in their study that Type I errors lead to higher costs for banks than to Type II error. They introduced a new performance measure, which they call weighted error, defined as the square root of the product of the model's accuracy, the percentage of bankrupt firms classified correctly, and the number of bankrupt firms to the total of predicted bankruptcies. Ghatasheh et al. (2020) assign different weights to Type I and Type II errors to reweight training inputs based on predetermined class cost. However, they still evaluate the models based on accuracy, Type I errors, Type II errors, G-mean, and AUC.

To the best of our knowledge, no other study so far has looked at the costs for the investor related to the debt level of the individual company, or the social costs of the models. We argue that taking a cost-sensitive perspective is crucial not only for the comparison, evaluation, and selection of different prediction models for bankruptcy, but also for an assessment of expected costs of the selected prediction model in terms of financial and/or socio-economic effects.

3. Methodology

3.1. Input variables

Since there is no unique set of variables used in previous prediction models, we consider three different sets of variables for our prediction models. These variable sets, based on accounting literature (Altman, 1968; Barboza et al., 2017; Ohlson, 1980) and shown in Table 2, are commonly used ratios for bankruptcy prediction models and are easily obtained from a company's balance sheet. Altman (1968) selected five ratios based on their statistical significance, evaluation of intercorrelations between the relevant variables, their predictive accuracy, and his judgement. Ohlson (1980) selected nine variables based on accounting literature. The Z-score and O-score are still some of the most used measures of bankruptcy probability today. The variables used by Barboza et al. (2017) are the original five variables from Altman, and six additional variables, which are used for organizational performance measuring (Carton & Hofer, 2006). Alfaro et al. (2008), Chen et al. (2010), Hillegeist et al. (2004), Jackson and Wood (2013), and Zelenkov and Volodarskiy (2021) also follow a similar approach, and select the input variables based on data availability and/or use in literature.

As the variables are commonly used in accounting literature, and, as shown in Fig. 3, not highly correlated, we believe that each of the included variables increases the predictive power, and used dimensionality reduction only as a robustness check.

All the variables are observed with a three-year delay, i.e., we use the value of these variables three years prior to the observed outcome (bankrupt or not bankrupt) in the predictions. We select this time delay so the companies have time to react and improve their financial health, as generating a bankruptcy prediction one or even two years prior the event would most likely be too late to prevent its consequences. As robustness checks, we also consider prediction models using a two-year delay (see Appendix).

3.2. Data description

We use financial data obtained from Compustat North America (Compustat, 2022) in this study. The data encompass bankruptcy events

Table 2
Variables used in the bankruptcy prediction models.

Based on Altman (1968)	Based on Barboza, Kimura and Altman (2017)	Based on Ohlson (1980)
X1 (liquidity) = *working capital/total assets*	X1 (liquidity) = *working capital/total assets*	SIZE = *log of total assets*
X2 (profitability) = *retained earnings/total assets*	X2 (profitability) = *retained earnings/total assets*	TLTA = *total liabilities/total assets*
X3 (productivity) = *EBIT/total assets*	X3 (productivity) = *EBIT/total assets*	WCTA = *working capital/total assets*
X4 (debt ratio) = *market value of equity/ total debt*	X4 (debt ratio) = *market value of equity/ total debt*	CLCA = *current liabilities/ current assets*
X5 (asset turnover) = *sales/assets*	X5 (asset turnover) = *sales/assets*	OENEG = *1 if total liabilities > total assets*
	OM (operational margin) = *EBIT/sales* GA (assets growth) = $(total\ assets_t - total\ assets_{t-1})/\ total\ assets_{t-1}$	NITA = *net income/total assets* CHIN = *change in net income*
	GS (sales growth) = $(total\ sales_t - total\ sales_{t-1})/\ total\ sales_{t-1}$	INTWO = *1 if the net income <0 during the last two years*
	GE (growth of employees) = $number\ of\ employees_t$- $number\ of\ employees_{t-1}/\ number\ of\ employees_{t-1}$	FUTL = *funds from operations/ total liabilities* (excluded because of the many missing values)
	CPB (change in price to book) = $price\ to\ book_t - price\ to\ book_{t-1}$ CROE (change in ROE) = $ROE_t - ROE_{t-1}$	

from 1985 to 2020, where data from 1985 to 2014 are used to train the different models and data from 2015 to 2020 are used as a separate test set. The overall dataset consists of 18,858 different companies and 193,027 company-year observations. During this period, 776 companies filed for Chapter 7 (liquidation) bankruptcy, the event that we focus on in this study. While this is an overall small number compared to the more prevalent Chapter 11 bankruptcies, using the latter would incur difficulties in quantifying the effects of the individual reorganization plans on the stakeholders. In a Chapter 11 bankruptcy event, the company restructures its debts and obligations but continues to operate. Therefore, it is difficult to estimate which impact this type of bankruptcy will have on the investors and employees, which is why we are excluding them from the sample. Overall, we can observe that the number of bankrupt companies per year was decreasing, and stayed especially low after the financial crisis of 2007 (Fig. 1). This increasing imbalance in outcomes makes it more difficult for the different models to classify the bankrupt companies correctly, which is why we use an undersampling strategy during model training. Specifically, we select all the observations from the minority class (Chapter 7 bankruptcy) and a random subset of the same size from the majority class in the training set. We use this method as the number of observations in the minority group is sufficiently large to create a balanced sample, so machine-learning methods can learn from it.

As it is common for financial data, more than 25% of the information is missing. Especially the financially distressed companies might have an incentive to hide their real status from their investors. Only 36% of the bankrupt companies published all the data used in this study for the year in which they filed for bankruptcy. In contrast, 57% of the non-bankrupt companies provide all the data. Gentry et al. (1985, p. 148) state that: “Frequently companies will stop reporting financial statements to Compustat two or more years before experiencing bankruptcy”. Out of the 28 bankrupt companies from the test set, only 15 provided information about the financial ratios used in this study. Different imputation methods lead to a reduced performance, and therefore only complete observations are included in the sample. The exclusion of the observations with missing data causes even higher imbalances in the test sample. Table 3 shows the number of bankrupt and non-bankrupt companies in the test set, depending on the input variables and time delay.

Almost all of the ratios used have some outliers on one or both sides of the distribution. We decided to leave them in the sample, as removing them would reduce the information about the variability of the ratios and could increase the imbalance of the test sample if some bankrupt companies were removed.

As we follow the approach of Barboza et al. (2017), we do not perform any data transformation, such as normalization. They state that: “Although this procedure may reduce the predictive power of the models, we aimed to analyze the adequacy of machine learning techniques without relying on specific or special treatment of data in the sample” (Barboza et al., 2017, p. 410).

Table 3
The test sets after removing incomplete observations.

Variables	Time delay	Bankrupt companies	Non-bankrupt companies
Altman	2 years	19	11088
Altman	3 years	18	11009
Altman plus	2 years	16	9472
Altman plus	3 years	15	9195
Ohlson	2 years	20	13822
Ohlson	3 years	20	13096

3.3. Prediction models

Similar to previous work in bankruptcy prediction, we apply a variety of different prediction models, including statistical prediction models such as Linear Discriminant Analysis and Logistic Regression and ML models such as Random Forests and Support Vector Machines. The following subsections provide a detailed description of the models used.

3.3.1. Linear discriminant analysis

Altman (1968) used the LDA to perform one of the first bankruptcy prediction analyses. The method was developed by Fisher (1936) in 1936 and is used to solve classification problems based on a linear combination of independent variables. The assumptions of LDA are a joint normal distribution and homoscedasticity, and it is sensitive to outliers. To distinguish between a bankrupt and non-bankrupt firm, it calculates a score based on the linear combination of the independent variables. Although other methods achieve better performance (Barboza et al., 2017), it is still used as a benchmark for other models. It is calculated as follows:

$$z = \sum_{i=1}^{n} (w_i x_i + c),$$

where z is the bankruptcy score, x_i the independent variable, c is a constant, and w_i the discriminant weight.

3.3.2. Logistic regression

Logistic regression is a conditional probability model that uses the non-linear maximum log-likelihood technique to estimate the probability of failure (Jackson & Wood, 2013). Ohlson (1980) was the first to use it for bankruptcy prediction in 1980. Over time, it became more popular than the LDA because of its less restrictive assumptions, and is considered to be the safer method in statistical literature (Veganzones & Séverin, 2018). It creates a score which is calculated as:

$$\mathbf{z} = \frac{\mathbf{1}}{\mathbf{1} + \mathbf{e}^{-(\mathbf{w_0} + \sum_{\mathbf{i}} (\mathbf{w_i x_i}))}},$$

where x_i are the independent variables, w_i the estimated weights, and z is the bankruptcy probability, represented with a value between 0 and 1.

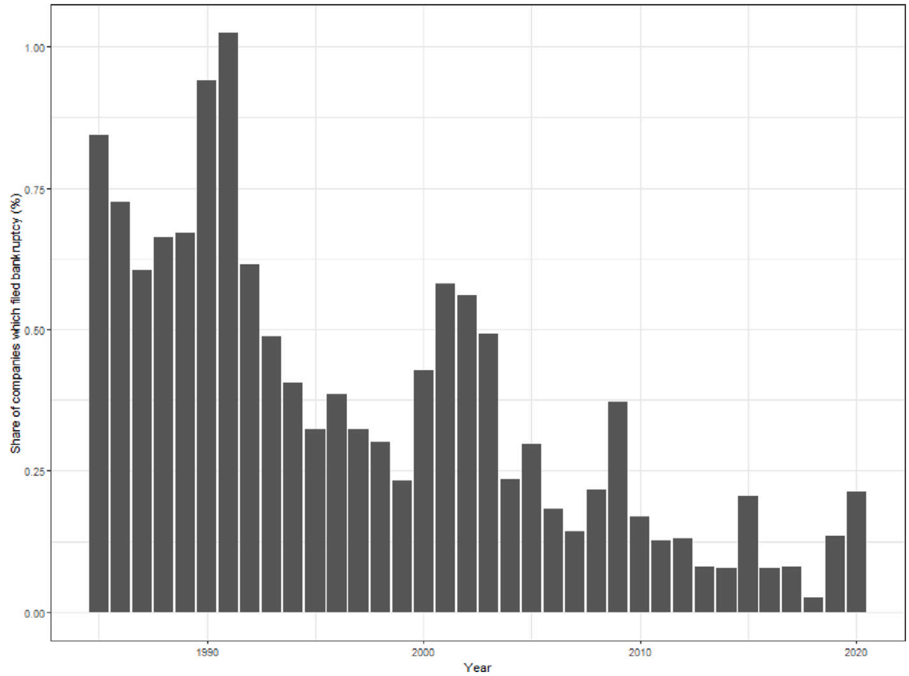


Fig. 1. Number of bankruptcies per year, focusing on the study period from 1985 to 2020.

3.3.3. Support vector machines

Support vector machines (SVMs) treat training examples as points in a high-dimensional space. Their basic idea is to construct a separating hyperplane with a high classification accuracy (Li et al., 2013), where observations falling on one side of the hyperplane are classified as one outcome, and observations falling on the opposite side as the opposite outcome. SVMs aim to find the hyperplane with the largest margin (distance to the nearest expression vector) while allowing some observations to fall on the 'wrong' side of the hyperplane (Noble, 2006). They were introduced by Cortes and Vapnik (1995) as a new learning machine for two-group classification problems. The optimization problem can be summarized as:

$$\underset{w,\xi,b}{\text{Minimise}}\ \frac{1}{2}w^T w + C\sum_{i=1}^{M}\xi_i,$$

$$\text{subject to } y_i[w^T\phi(x_i)+b] \geq 1-\xi_i,$$

where C are the classification costs, $i \in 0,1,2...,M$ are the training observations, $\xi_i \geq 0$ are the margins of error, y_i are the classifications in the training set, and $\phi(x)$ is the function which transforms the space $\mathbf{R}^M$. The function $\phi(x)$ does not need to be known, as a kernel function is applied so that $K(x) = \phi(x_i)^T \cdot \phi(x_j)$ (Li et al., 2013).

SVMs are widely used for bankruptcy prediction problems due to their advantages compared to other machine-learning approaches (Shin et al., 2005), including the fact that they only have two free parameters (the classification costs and the Kernel function), the guarantee that an optimal, global solution exists, and their general predictive performance.

3.3.4. Boosted decision tree

Boosting is a machine-learning method that can be used for regression and classification tasks. It is used for bankruptcy prediction, as it is considered one of the best-performing methods in machine learning (Qian et al., 2022). It sequentially creates ensemble models based on weak learners, usually decision trees. It assigns every observation an equal weight and, after the classification is performed, the weight of the incorrectly classified observations is increased in the next step. This process is repeated until a stopping criterion is reached (e.g. the number of iterations) (Freund, 1995). We use a gradient boosting algorithm, which minimizes the loss function of the learner (Friedman, 2002), and follow the approach of Qian et al. (2022):

1. Initialize a model with a constant value $F_0(x) = \arg\min_\gamma \sum_{i=1}^{n} L(y_i,\gamma)$,
2. For each iteration $m = 1,\dots,M$ do:
   (a) Compute pseudo-residuals $r_{im} = -[\frac{\delta L(y_i,F(x_i))}{\delta F(x_i)}]_{F(x)=F_{m-1}(x)}$ for $i = 1,2,\dots,n$,
   (b) Fit a base learner $h_m(x)$ to pseudo-residuals,
   (c) Compute the multiplier γ_m by solving the optimization problem $\gamma_m = \arg\min_\gamma \sum_{i=1}^{n} L(y_i, F_{m-1}(x_i)+\gamma h_m(x_i))$,
   (d) Update the model: $F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$.

$L(y,F(x))$ represents the loss function, $(x_i,y_i)_{i=1}^n$ the training set, and M is the number of iterations.

3.3.5. Random forest

The random forest algorithm was created by Breiman (2001) in 2001. Random forests are ensembles of (independent) decision trees and usually outperform decision trees because they overcome the overfitting problem. Other benefits of random forests are that they are robust and allow outliers and noise in the data (Yeh et al., 2014). Random forests are created as a collection of independent trees, and the random forest algorithm grows each tree by randomly selecting a subset of features from each node of the tree at each step. Each individual tree predicts the outcome for an observation, and the class is selected based on majority voting or weighted probability voting (Barboza et al., 2017). Our approach is similar to Barboza et al. (2017):

1. Create random subsets of the data set, with an arbitrary number of observations and different features.
2. Each subset generates one decision tree, and all elements of the set have a label.
3. For each observation, the forest generates a large number of predictions (votes), one for each tree. The observation is assigned to the class with the most votes.

Table 4
The hyperparameters for machine-learning models.

Type	Model	Hyperparameter	Values	R package
Statistical	Logistic regression	–	–	–
	Linear discriminant analysis	–	–	–
ML	SVM	sigma	0.01, 0.015, 0.2	svmRadial
		c	0.505, 0.9, 1, 1.1, 1.25	
	Boosted decision tree	Interaction depth	1, 3, 5	gbm
		Number of trees	50:2500	
		Shrinkage	0.01, 0.001	
	Random forest	Correlation of trees (mtry)	1:10	rf
	Neural network	Size	5, 10, 15, 20, 25	nnet
		Decay	0, 0.1, 0.001, 0.0001, 1	
	Bagged decision tree	Max. depth	1:10	AdaBag
		Number of trees	50, 100, 150	

3.3.6. Neural network

Neural networks are one of the most widely used machine-learning approaches. A neural network is a mathematical function that imitates the biological neural networks of human brains. They consist of nodes called neurons, which are connected and able to transmit signals to other neurons. Neural networks transform all data into a set of numerical vectors, which are used as input for the network. Each neuron receives some input and creates output, which can be sent to one or multiple other neurons. Each connection has a weight, and the output is the weighted sum of all inputs. The network consists of layers; the layer which receives the input is called the input layer, the one which creates the output is called the output layer. Between them, there can be one or multiple hidden layers. An non-linear activation function is used to decide if a node will be activated. Messier and Hansen (1988) were the first who used neural networks for the purpose of bankruptcy prediction. Their popularity increased over time; out of the 10 most cited bankruptcy-prediction papers, according to Shi and Li (2019), 6 use neural networks. Besides the high performance, neural networks also have some disadvantages (Zhao et al., 2015): they show a poor performance when an unbalanced training set is used, and it is difficult to select the optimal number of layers. We use a simple, single-layer neural network in this study.

3.3.7. Bagged decision tree

Bagging, or bootstrap aggregating, was created by Breiman (1996). It is performed by creating random subsamples with replacement from the initial data set, and by creating as many models as there are samples. Any classification or regression technique can be used (e.g. decision trees). The classification is performed by majority voting. The purpose of the recombination of the training set is to reduce overfitting. Our approach is based on Breiman (1996) and Barboza et al. (2017) and follows the following steps:

1. A random subset $t = 1$, is selected from the data set,
2. Classifiers c_t are configured based on the subset,
3. The previous two steps are repeated for $t = 2, \ldots, T$,
4. Each classifier determines a vote, $C_t(X)$, where x is the data of each element from the training set.
5. At the end, the class with the largest number of votes is selected according to $C(x) = T^{-1} \sum_{t=1}^{T} C_t(x)$,

3.4. Model training and model evaluation

To properly fit the machine-learning models to our scenario, we use grid search hyperparameter optimization and a 10-times repeated 10-fold cross-validation during model training. To select the best hyperparameters, each individual model is trained on the balanced test sets. Table 4 lists the prediction models considered and their corresponding hyperparameters. We consider standard hyperparameter optimization based on statistical measures, specifically the accuracy.[1] The accuracy of the model is calculated as the number of correct classifications divided by the total number of observations. For all the models, without loss of generality and due to the undersampling strategy, a default threshold of 0.5 is used to divide the predicted probabilities into classes, where a predicted probability of ≥ 0.5 gets classified as 'bankrupt'.

As only the training folds are balanced through undersampling yet the test sample is not, we use balanced accuracy as the statistical evaluation measure since accuracy is not an adequate measure for imbalanced datasets. This is because the impact of the class with the lowest number of observations is reduced, compared to the majority class (Branco et al., 2016). As bankruptcy, the least common value of the response variable, is the most important one, the evaluation metrics should consider the distribution of the data. The balanced accuracy is the average of the accuracy rates for bankrupt and non-bankrupt companies. We also calculate the sensitivity (true positive rate), specificity (true negative rate) and area under the ROC curve (AUC).

To investigate the financial costs and implications of bankruptcy predictions, we calculate the saved costs based on Altman et al. (1977), who approximates the costs of a Type I error as 70% of the total debt at the time of bankruptcy, and the costs of a Type II error as 2% of the amount of money that could be lent. The Type I error is calculated as the false negative rate and the Type II error as the false positive rate:

Type I error = FN / (TP + FN)

Type II error = FP / (TN + FP)

These estimations are based on an empirical analysis of annual report data of the 26 largest US commercial banks. However, the LGD shows a large variance, and the recovery rates are smaller for larger companies and companies that fail on multiple loans. As our dataset consists of observations over 35 years and does not provide data about the loan recovery rates, it is difficult to estimate the LGD. Therefore, we use the costs based on Altman et al. (1977) as a starting point of the analysis and perform robustness checks with various other values of Type I and Type II errors. The total debt variable is adjusted for inflation and all the values are in 2020 USD.

Overall, the cost reduction is calculated as the difference between the costs of a naïve model, where every company receives a loan, and the costs of the prediction model, where the binary predictions are used for the loan decision. To be precise, if the predicted outcome is bankrupt, no loan would be given to this company, whereas a prediction of not-bankrupt leads to a loan. From a notation perspective, we consider B as the set of bankrupt companies, B_{FN} as the set of bankrupt companies that the prediction model incorrectly classified as solvent, and B_{FP} as the set of solvent companies that the prediction model incorrectly classified as bankrupt. We then define the different cost components as follows:

[1] Using accuracy for hyperparameter optimization is possible since we use undersampling to create balanced training folds.

$$TEC\ (true\ error\ costs) = 0.7 * \sum_{i \in B} Total\ debt_i,$$

where B is the set of all bankrupt companies from the dataset.

$$FNC\ (false\ negative\ cost) = 0.7 * \sum_{i \in B_{FN}} Total\ debt_i,$$

where B_{FN} is the set of bankrupt companies which the model classified as not bankrupt.

$$FPC\ (false\ positive\ cost) = 0.7 * \sum_{i \in B_{FP}} Total\ debt_i,$$

where B_{FP} is the set of not bankrupt companies which the model classified as bankrupt.

$$\boldsymbol{Cost\ reduction = TEC - (FNE + FPE)}$$

$$\boldsymbol{Cost\ reduction}(\%) = \frac{\mathbf{TEC - (FNE + FPE)}}{\mathbf{TEC}}$$

Specifically, while the cost reduction measures the absolute reduction of costs from a prediction model, compared to a naive model where all companies get a loan, we also calculate the percentage cost reduction to make a comparison across models easier.

We also use the unpredicted job loss as a proxy for the social impact of the model. We calculate this as the number of employees of the bankrupt companies which were classified as solvent by the model yet filed for bankruptcy (companies in set B_{FN}). The affected employees were not able to anticipate the failure of their company and are therefore affected more than the employees of the correctly classified companies. As we observe only companies which filed for Chapter 7 bankruptcy (liquidation) and do not include companies which filed for reorganization, we assume that all the employees of the bankrupt companies lose their jobs.

$$\textbf{UJL (unpredicted job loss)} = \sum_{i \in B_{FN}} Number\ of\ employees_i,$$

where B_{FN} is the set of all bankrupt companies which the model classified as solvent.

$$\textbf{UJL}(\%) = \frac{\sum_{i \in B_{FN}} Number\ of\ employees_i,}{\sum_{i \in B} Number\ of\ employees_i,}$$

To make the results more comparable across different models, we divide the UJL with the total number of employees in bankrupt companies and use the percentage of employees affected by the unexpected job loss as the measure for the social impact.

4. Exploratory data analysis

Before we train and evaluate the different prediction models, we start with an exploratory analysis of the data and bankruptcy events. During the period between 2015 and 2020 (the test set), 28 companies went bankrupt. Out of these companies, 78.6% received a loan during the last three years of their existence. Some of them increased their debt significantly during the two years before they filed for bankruptcy. Out of the three bankrupt companies with the highest debt, two received a loan two years before bankruptcy, and one in the year before it filed for bankruptcy. This shows that many investors underestimated the risk of the investment and are not able to recognize the companies' financial distress. Figs. 2(a) and 2(b) show that the bankrupt companies have some extreme outliers regarding the amount of debt and number of employees. Therefore, we can conclude that the misclassification of a bankrupt company would not have the same effect for every observation in the sample. Misclassifying a larger company would likely lead to more severe consequences than misclassifying one with a low amount of total debt or number of employees.

Fig. 3 shows that the ratios that are used in the same model are not highly mutually correlated, and therefore are suitable as input variables for the logistic regression and discriminant analysis, as well as all machine-learning approaches. Also, none of the independent variables are highly correlated with the dependent variable.

Table 5
The performance of machine-learning models with different input variables and a 3-year delay (SENS — sensitivity, SPEC — specificity, BACC — balanced accuracy, AUC — area under the ROC curve, CR — cost reduction, UJL — unexpected job loss).

Model	SENS	SPEC	BACC	AUC	UJL (%)	CR (%)
Altman variables						
LDA	0.89	0.47	0.68	0.73	7	−32
Logistic regression	0.89	0.47	0.68	0.73	7	−32
Random forest	0.72	0.73	0.73	0.83	23	40
Neural network	0.67	0.78	0.71	0.83	22	40
Bagging	0.72	0.75	0.74	0.83	9	43
Boosting	0.78	0.77	0.78	0.86	9	48
SVM radial	1.00	0.53	0.77	0.81	0	6
Altman plus variables						
LDA	0.73	0.65	0.69	0.72	7	5
Logistic regression	0.80	0.62	0.71	0.75	7	6
Random forest	0.80	0.77	0.78	0.87	16	53
Neural network	0.80	0.76	0.78	0.80	9	53
Bagging	0.73	0.82	0.77	0.88	16	55
Boosting	0.87	0.74	0.80	0.86	15	47
SVM radial	0.87	0.64	0.75	0.84	33	35
Ohlson variables						
LDA	0.70	0.59	0.64	0.74	35	−53
Logistic regression	0.70	0.60	0.65	0.74	35	−49
Random forest	0.90	0.66	0.78	0.81	16	38
Neural network	0.90	0.69	0.79	0.82	16	48
Bagging	0.90	0.63	0.76	0.81	16	36
Boosting	0.90	0.65	0.77	0.83	16	36
SVM radial	0.85	0.66	0.76	0.81	18	46

Fig. 4 shows the average value of each variable for bankrupt and non-bankrupt companies. The stars above the bars represent the significance of a t-test that compares the two groups (bankrupt vs non-bankrupt). Solvent companies on average have a statistically significant increase in assets and sales, while bankrupt companies experience a decrease. Bankrupt companies also show a lower net income to total assets ratio, working capital ratio, retained earnings to total assets ratio, and a lower equity to debt ratio. Bankrupt companies also often have a total liabilities to total assets ratio which is larger than one, and a negative income during the previous two years.

5. Evaluation

5.1. Bankruptcy prediction and financial impact/costs

Table 5 shows the performance of the different prediction models on the test set, based on various evaluation metrics. Overall, the results clearly show that machine-learning models are superior compared to logistic regression or discriminant analysis in terms of AUC, balanced accuracy (BACC), cost reduction (CR) and unpredicted job loss (UJL). The percentage of the unpredicted job loss is calculated as the share of the jobs lost because of the unpredicted bankruptcies divided by the total number of jobs lost. Similarly, the percentage of the cost reduction is calculated as the ratio of the costs of a model and the costs of a naive model, where every company in the dataset receives a loan.

If the companies are compared based on the number of employees affected, the results are mixed. For the Altman and Altman plus variables, the statistical models achieve a better performance than many ML models. This can be explained by the fact that those models achieve a higher sensitivity and therefore classify more of the bankrupt companies correctly. However, they also generate more false positives, which reduces their overall accuracy and cost reduction.

The results show that the variance of the cost reduction and unpredicted job losses is much larger than the variance of the other metrics.

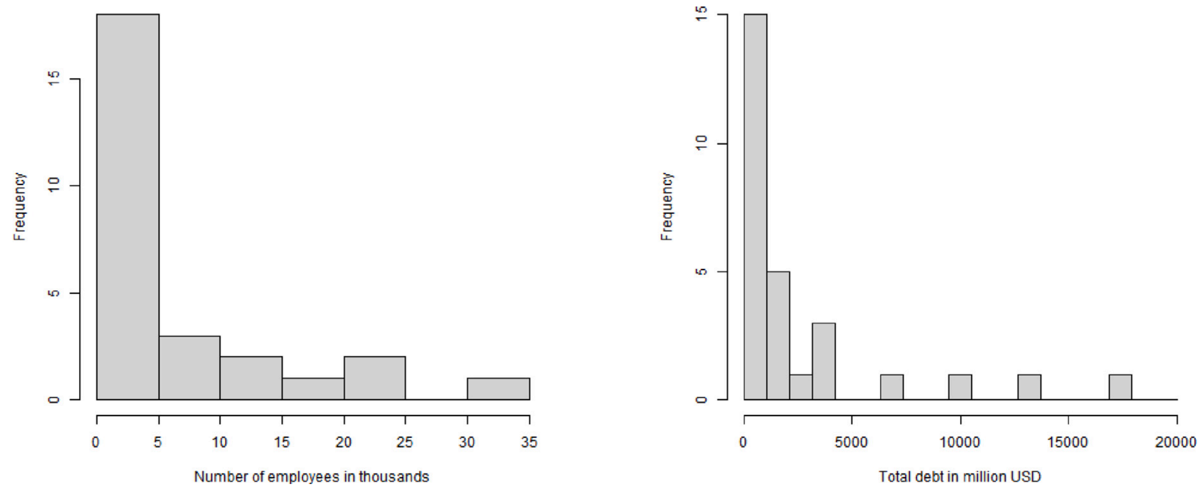


Fig. 2. The distribution of the size of the bankrupt companies.

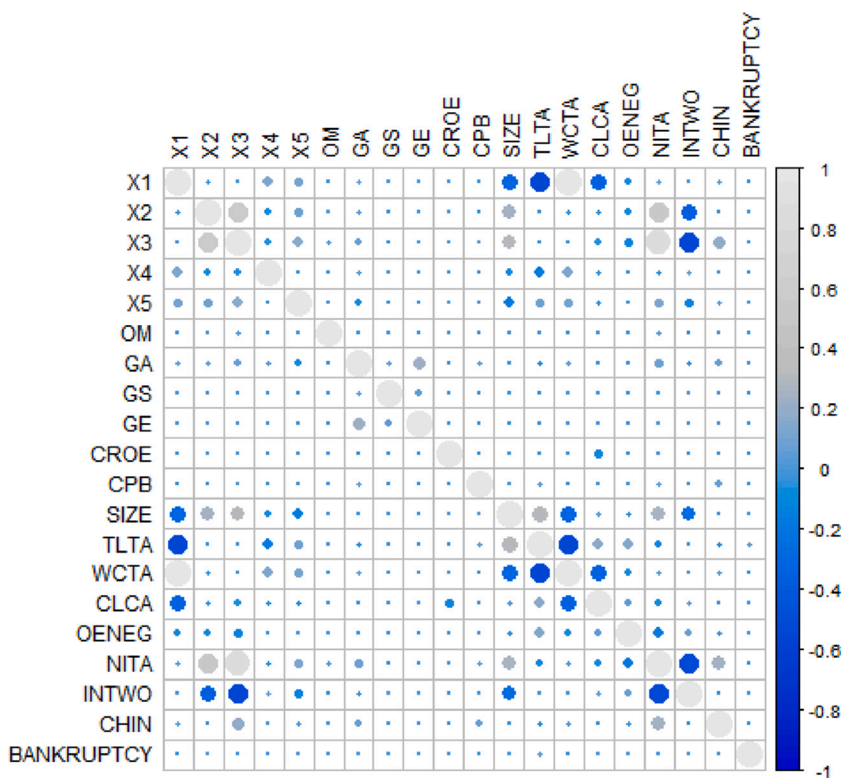


Fig. 3. Correlation plot of the considered predictor variables. See Table 2 for their detailed definition.

The value for AUC ranges from 0.72 for the discriminant analysis with the Altman plus variables to 0.88 for the bagging model with the same input variables. The balanced accuracy varies from 68% to 80%. However, the profit is in the range from a 53% decrease to a 63% increase, and the unpredicted job loss varies from 0% to 35%. The statistical models, which also perform worse regarding other performance metrics, generate a negative profit for two sets of input variables and significantly lower profit than ML models with the Altman plus variables.

A surprising result is that models with a similar AUC can have a large difference in cost reduction. For example, for the models with the Altman variables, the neural bagging an AUC of 0.88, and SVM has an AUC of 0.84. However, the profit increase generated by these models is 55% and 35%, respectively. This difference leads to USD 8,043 million of saved costs for the investors. This shows that integrating financial evaluation metrics that measure the real-world financial and socio-economic impact of bankruptcy prediction models is crucial and has a significant effect on the selection of an appropriate prediction approach. Furthermore, models with similar AUC can also generate significantly different unexpected job losses. With the Altman variables, the random forest and bagging models both have an AUC of 0.83, but the values of the UJL for those models are 23% and 9%, respectively. This difference translates into over 11,000 employees affected by the unpredicted bankruptcy.

Fig. 5 shows the mean AUC, BACC, cost reduction and unpredicted job loss of the machine-learning and statistical models on the three sets of input variables, ordered by AUC. The Altman plus variables result in the highest AUC, BACC and cost reduction if ML models are applied. The AUC for the ML models ranges from 0.86 for the Altman plus variables to 0.81 for the Ohlson variables, while the cost reduction

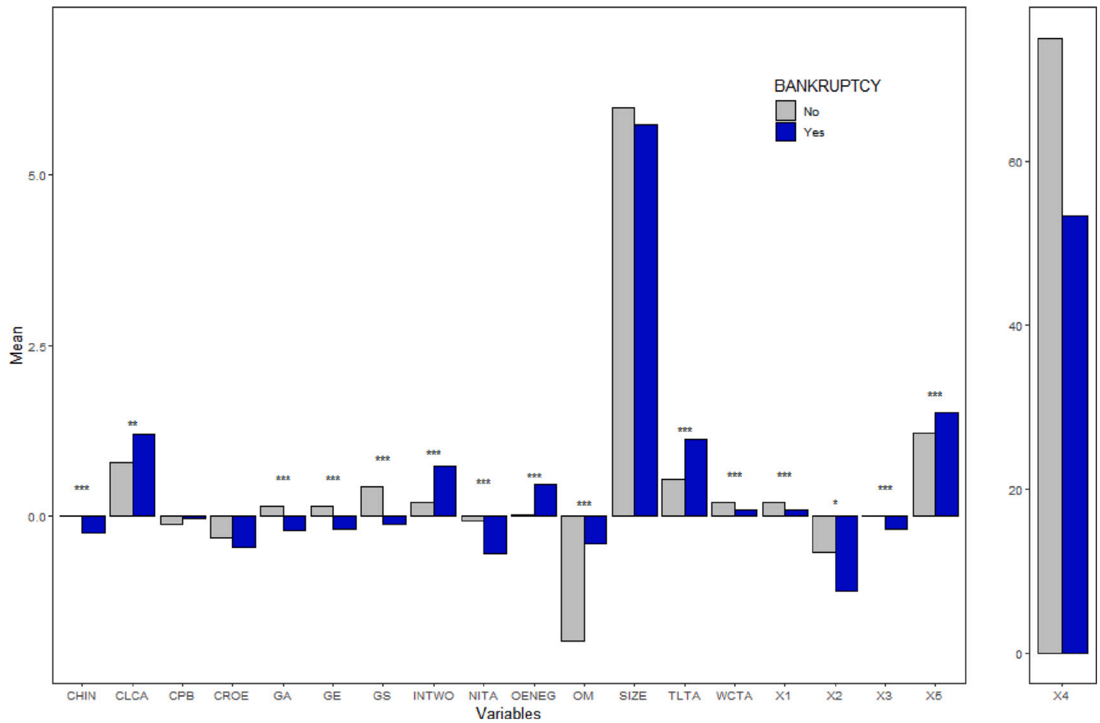


Fig. 4. The mean value of the explanatory variables grouped by bankruptcy. The labels represent the significance of the t-test ($p < 0.05$ *, $p < 0.01$ **, $p < 0.001$ ***).

has a larger variance and ranges from 46% to 31%, respectively. The Altman variables also account for the lowest average unpredicted job loss of 11%, while the Altman plus and Ohlson variables perform worse (16% and 12%, respectively). If the statistical methods are applied, only the Altman plus variables lead to a cost reduction, while the other two sets of variables lead to increased costs. The Ohlson variables, although having the largest AUC, generate the largest cost increase and unpredicted job loss. The Altman plus variables are the only group of variables which can generate a cost reduction even when with statistical models.

Fig. 6 shows the average AUC, BACC, cost reduction and unpredicted job loss of the models on the three sets of input variables, ordered by AUC. The statistical models on average generate a cost increase, while the machine-learning models all generate a cost reduction. Neural networks, which are ranked fifth based on AUC, cause the largest average cost reduction. The random forest and bagging models, which are ranked third and second based on AUC, respectively, have a similar performance regarding AUC and cost reduction, but the unpredicted job loss is 50% higher for the random forest model.

5.2. Robustness checks

We tested the robustness of our predictions with different alternative scenarios for the financial effects of the different prediction outcomes. In Fig. 7, we show the average performance of the predictions if we use a 2-year instead of a 3-year delay, grouped by model and input variables. The results are consistent with our main findings. When ML models are applied, the Altman plus variables result in the largest AUC, balanced accuracy, cost reduction and lowest unpredicted job loss. The Altman variables generate a slightly higher unpredicted job loss and a significantly higher cost reduction than the Ohlson variables. The statistical models again perform worse than the machine-learning models regarding all performance measures.

Fig. 8 shows the results of our analysis with a 2-year delay grouped by model. Although the ranking of the models is different compared to the results with a 3-year delay, we observe again that the ranking of the models depends on the performance measure. For example, the random forest model is ranked forth regarding AUC, but leads to the largest cost reduction. The boosting model, which is ranked first regarding AUC and BACC, and the statistical models again show the worst performance regarding all four metrics.

We also performed robustness checks and calculate the cost reduction with different values for the Type I and Type II errors. According to the Basel II accord (de Basilea, 2006), the loss given default is between 45% and 75% of the total debt, depending on the type of loan and collateral. As we have no further information about the loans, we performed our analysis with both of those values, and additionally with a Type I error equal to their mean — 60%.

When approximating the costs of not lending to a company that will remain solvent, we rely on the market risk premium. As the investors always have the alternative to invest in risk-free government bonds, they lose only the difference between the interest on the government bonds and the interest they would have received from the loan. As the risk premium on lending in the USA was between 2.4% and 3.6% for the observed period (World Bank, 2022), we performed our calculations with 2.5%, 3% and 3.5% of the total debt as the costs of the Type II error.

The results of the robustness checks for different Type I and Type II error costs are shown in Figs. 9 and 10. The models are all ranked based on descending AUC. The figures show that our results stay consistent, regardless of which combination of Type I and Type II error costs we use. The ranking of the models based on cost reduction is not the same as the ranking based on AUC, and the variance of the cost reduction stays much larger than the variance of the traditional performance measures. When the results are grouped by input variables and method (Fig. 9), the Altman plus variables show the best average results regarding costs with both machine-learning and statistical models. When statistical models are applied, the Ohlson variables always show the largest cost increase, although they perform best regarding AUC.

If the results are grouped by model, the neural network, which is ranked fourth regarding AUC, is the best-performing model regarding cost reduction with all combinations of different Type I and Type II error costs (Fig. 10). Even when the Type II error costs are reduced to 45% of the total debt, it is the only model which does not lead to a cost

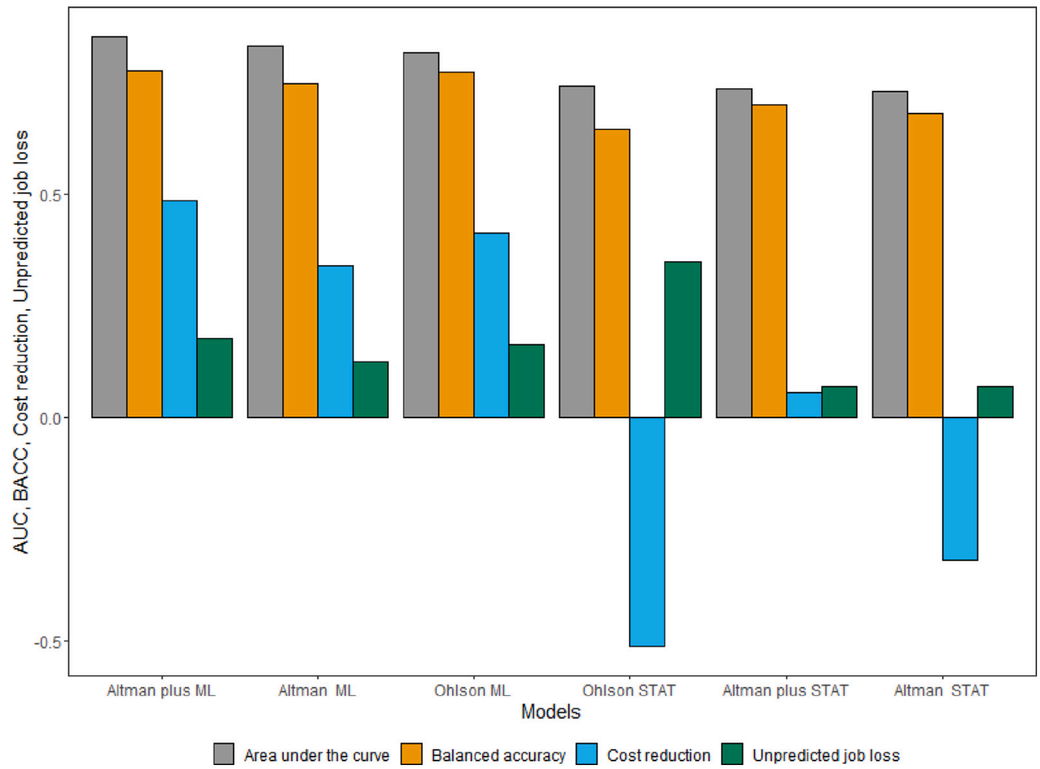


Fig. 5. The mean of the different models' AUC, BACC, cost reduction and unpredicted job loss on all three sets of input variables with a 3-year delay (ML — all machine-learning models, STAT — logistic regression and discriminant analysis).

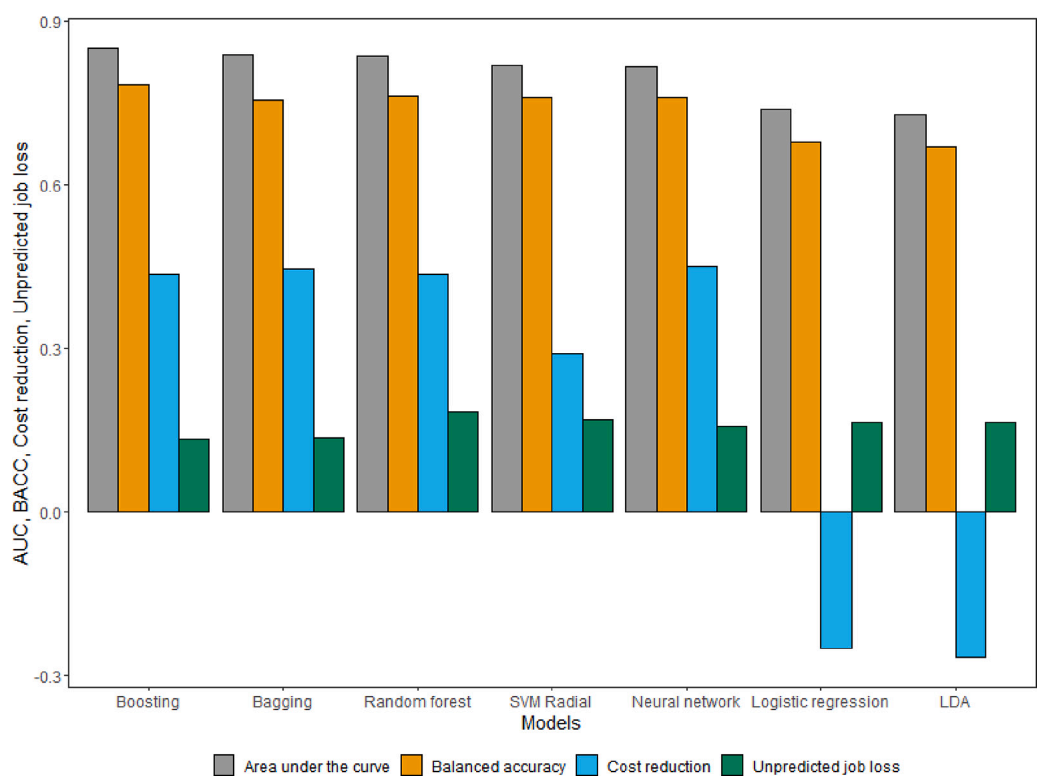


Fig. 6. The mean of the different models' AUC, BACC, cost reduction and unpredicted job loss on all three sets of input variables with a 3-year delay (grouped by model).

increase for any value of the Type I error. The boosting model, which performs best regarding AUC and BACC, is most often ranked second or third regarding cost reduction with the different Type I and Type II error costs. The statistical models always lead to a cost increase.

As we use a small neural network in the main analysis, we also performed robustness checks with a more complex recurrent neural network (RNN) architecture. After tuning its hyperparameters (e.g., learning rate, batch size, number of epochs), we applied the RNN with

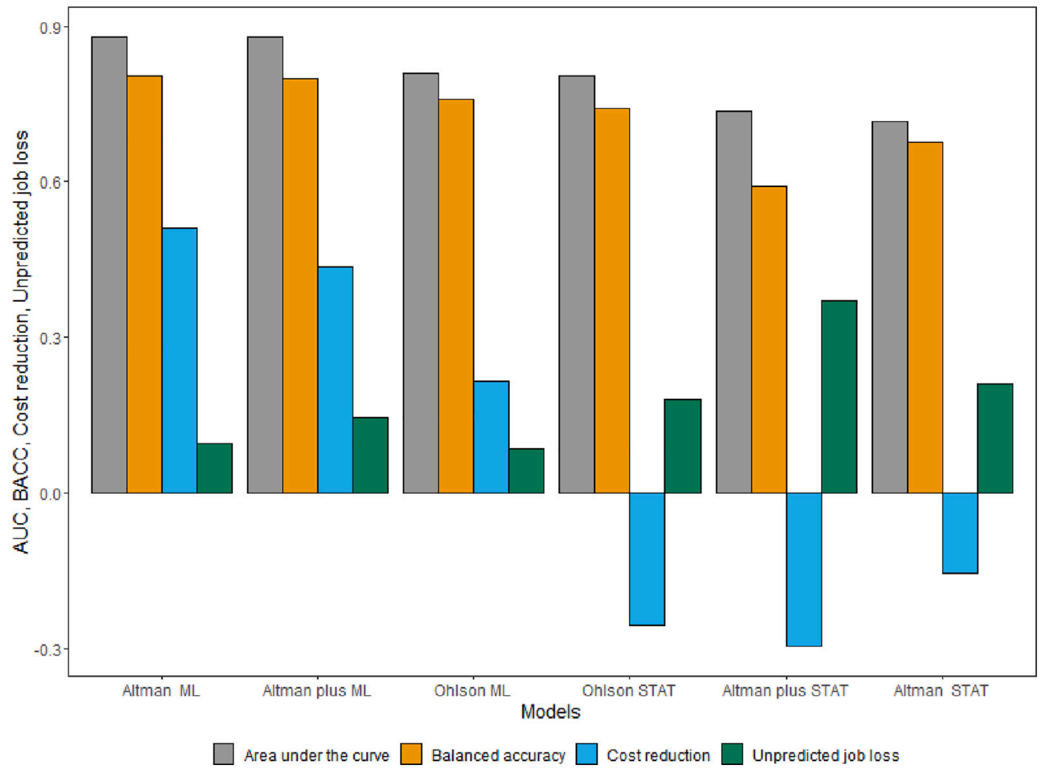


Fig. 7. The mean of the different models' AUC, BACC, cost reduction and unpredicted job loss on all three sets of input variables with a 2-year delay (ML — all machine-learning models, STAT — logistic regression and discriminant analysis).

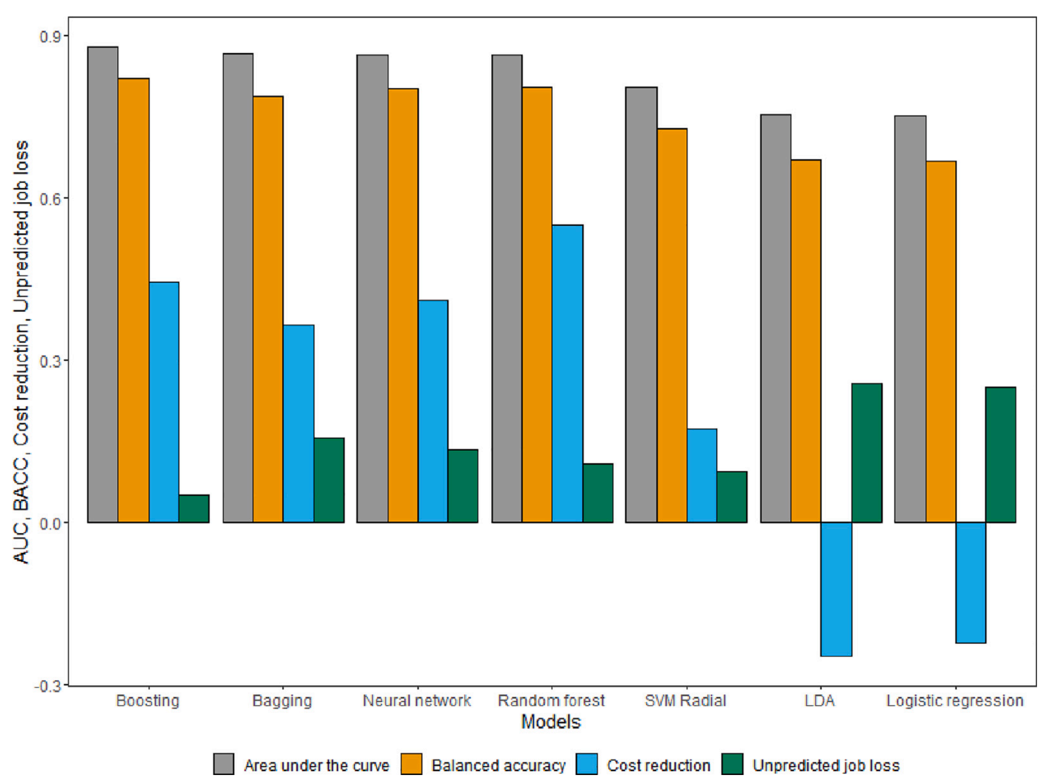
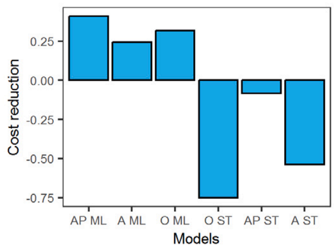


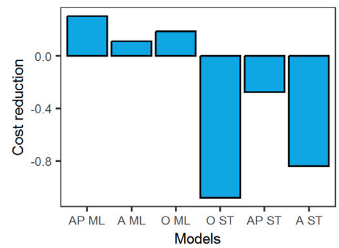
Fig. 8. The mean of the different models' AUC, BACC, cost reduction and unpredicted job loss on all three sets of input variables with a 2-year delay (grouped by model).

10 hidden features and calculated its AUC, BACC, and cost reduction performance. Fig. 11 shows that the Altman and Altman plus variables have a similar AUC, and all three sets of input variables have a similar balanced accuracy. However, the difference in the cost reduction is much larger between the different sets of variables, indicating that the RNN with Altman variables yields greater cost reductions.

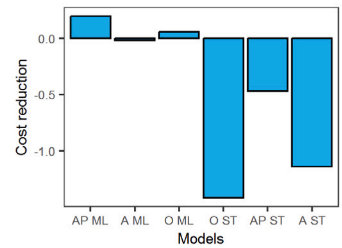
We also performed additional tests with an alternative dataset: Compustat data from companies in the European Union. Fig. 12 shows



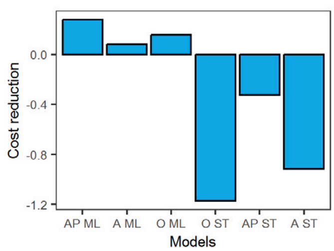
(a) Type I error = 0.025, Type II error = 0.75



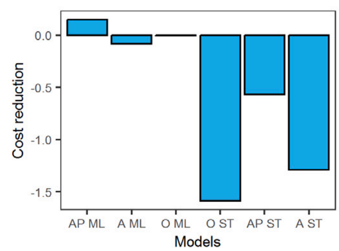
(b) Type I error = 0.03, Type II error = 0.75



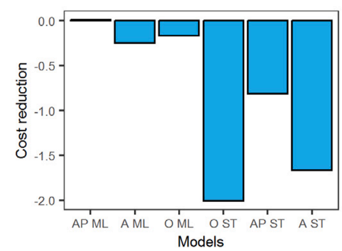
(c) Type I error = 0.035, Type II error = 0.75



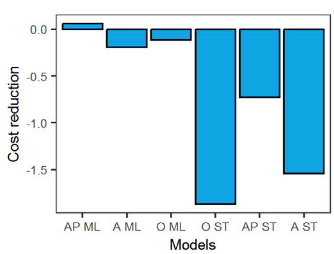
(d) Type I error = 0.025, Type II error = 0.6



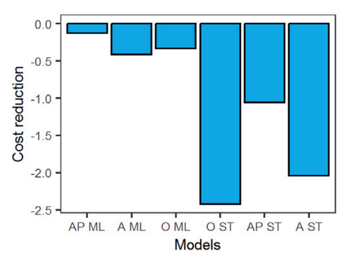
(e) Type I error = 0.03, Type II error = 0.6



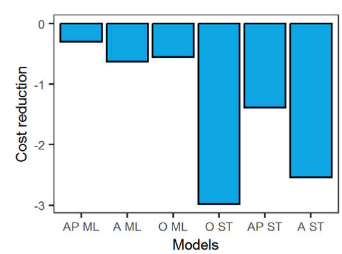
(f) Type I error = 0.035, Type II error = 0.6



(g) Type I error = 0.025, Type II error = 0.45



(h) Type I error = 0.03, Type II error = 0.45



(i) Type I error = 0.035, Type II error = 0.45

Fig. 9. The mean of the different models' AUC, BACC, CR and UJL on all three sets of input variables with a 3-year delay (ML — all machine-learning models, STAT — logistic regression and discriminant analysis, AP — Altman pus variables, A — Altman variables, O — Ohlson variables).

the performance of the different models on this dataset. As the dataset is much smaller and more imbalanced than the one from North America, all the models lead to a cost increase. However, our main findings are still consistent — the variance with respect to financial performance is much higher than the variance of the statistical performance, and the ranking of the different models depends on the selected evaluation metric.

Our results also stay consistent when the models are applied on the Altman plus variables with a reduced number of features. For the feature selection step, we applied a least absolute shrinkage and selection operator (LASSO) approach, selected the seven most important variables (X4, X1, X3, X5, OM, GS and CPB) and applied all the models on the reduced dataset. This robustness check also confirms our results — the cost reduction and unpredicted job loss show a larger variance than the balanced accuracy and the AUC (see Fig. 13).

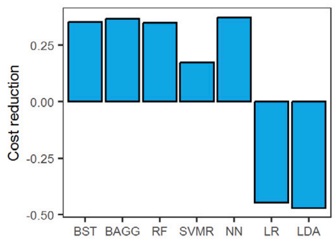
Figs. A.1 and A.2 in the Appendix show additional robustness checks with different Type I and Type II error costs when a two-year delay instead of a three-year delay. They confirm our main results. Similar to the analysis with a three-year delay, machine-learning models outperform the statistical models regarding cost reduction, and the cost reduction shows a much higher variance than the traditional performance measures for all combinations of Type I and Type II error costs.

6. Discussion

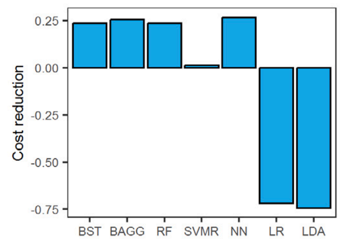
Across the different scenarios considered, we see that ML prediction models systematically outperform more traditional statistical prediction models such as Logistic Regression and Linear Discriminant Analysis. Furthermore, the results show that small differences in traditional evaluation metrics such as (balanced) accuracy and AUC can lead to substantial differences in financial costs and number of affected employees of the resulting predictions. In other words, selecting bankruptcy prediction models solely based on classification performance without taking the financial and socio-economic impacts into account could lead to a suboptimal selection of the prediction model.

6.1. Limitations

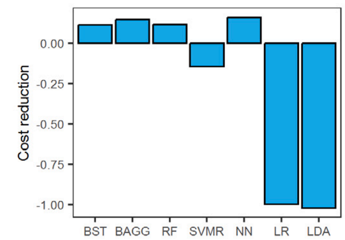
This study has several limitations: We selected the samples based on data availability and only include complete observations. Missing accounting information is more common in the balance sheets of bankrupt than non-bankrupt companies, as financially distressed companies may have interest in not reporting their accounting data, as they do not want to reveal their status. Therefore, such a data sample might be biased. One other problem is the "snapshot character of the annual account" (Balcaen & Ooghe, 2006, p. 77). As many accounting variables are available only on an annual basis, they might lead to incorrect conclusions. It is possible that a solvent company



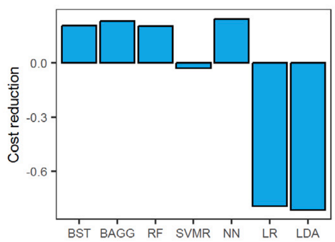
(a) Type I error = 0.025, Type II error = 0.75



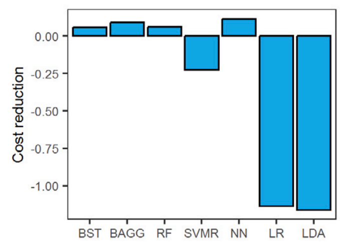
(b) Type I error = 0.03, Type II error = 0.75



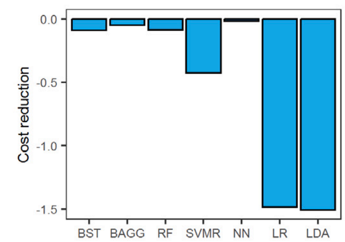
(c) Type I error = 0.035, Type II error = 0.75



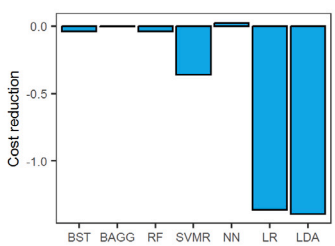
(d) Type I error = 0.025, Type II error = 0.6



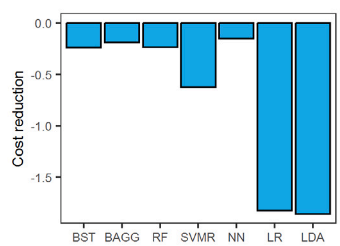
(e) Type I error = 0.03, Type II error = 0.6



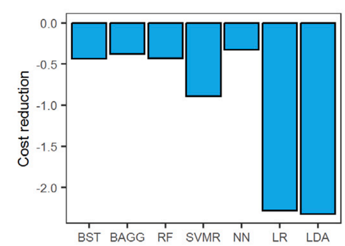
(f) Type I error = 0.035, Type II error = 0.6



(g) Type I error = 0.025, Type II error = 0.45



(h) Type I error = 0.03, Type II error = 0.45



(i) Type I error = 0.035, Type II error = 0.45

Fig. 10. The mean of the different models' AUC, BACC, CR and UJL on all three sets of input variables with a 3-year delay (BST — Boosting, BAGG — Bagging, RF — random forest, NN — neural network, LR — logistic regression, LDA — linear discriminant analysis).

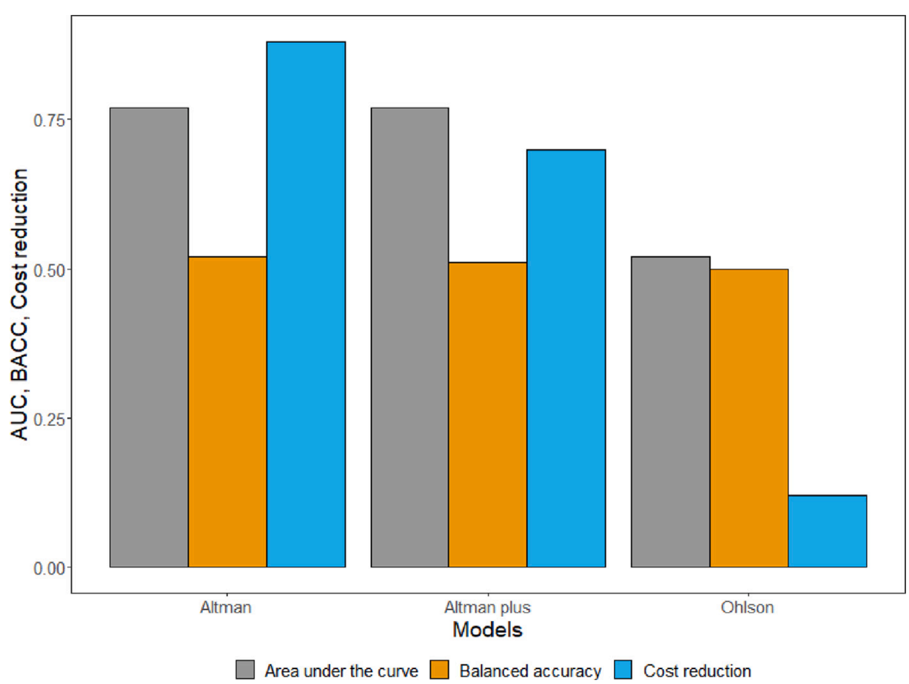


Fig. 11. The AUC, BACC and cost reduction of the RNN based on three previous years on all three sets of input variables.

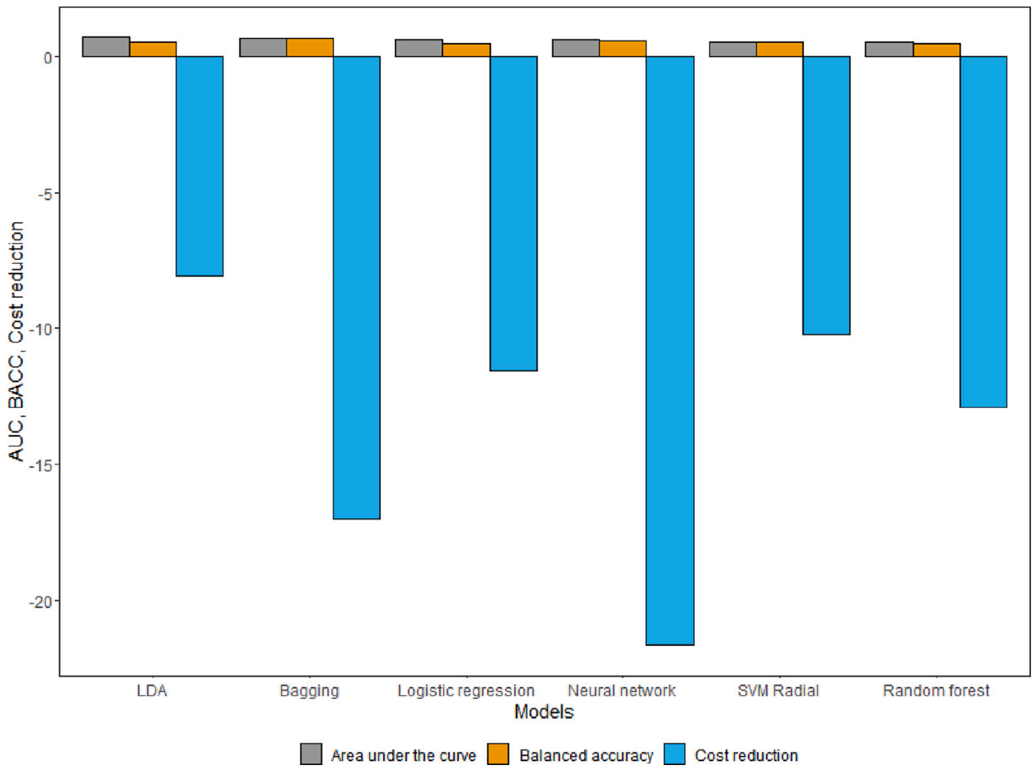


Fig. 12. The mean of the different models' AUC, BACC and cost reduction loss on all three sets of input variables with a 3-year delay on the European dataset (grouped by model).

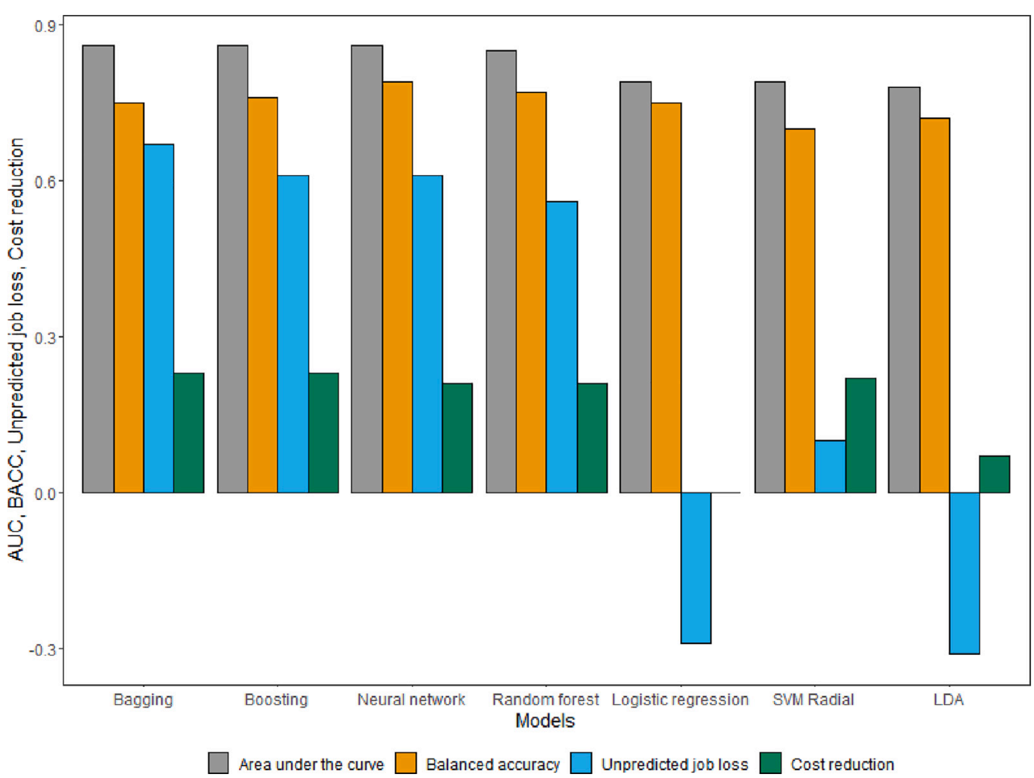


Fig. 13. The mean of the different models' AUC, BACC, cost reduction and unpredicted job loss on the Altman plus variables with reduced dimensionality, with a 3-year delay.

with temporary difficulties is classified as bankrupt. One limitation is also that the models use only financial data. It is not certain how other data would influence the accuracy metrics of the model. It has been proven that the models lose accuracy when they are used for other periods (Grice & Dugan, 2001). Financial data cannot incorporate proxies for non-financial events that can cause bankruptcy, like lawsuits and union problems. Other variables, which have been proven useful in bankruptcy prediction, such as different corporate governance indicators, macroeconomic variables, or textual data from the companies' reports, are not considered in this analysis. This leaves some space for future research to investigate if a combination of different variables could lead to improved predictions. One final limitation is also that the impact of a Type II error on the company cannot be evaluated. Denying a loan to a solvent company might not cause job losses, but it could reduce the profitability of the company and make its growth more difficult. On the other hand, denying a loan to a company which is financially distressed might prevent the company from pursuing a project which would increase the income. Future research should focus more on the different impacts of Type I and Type II errors on the company.

6.2. Summary and outlook

Predicting bankruptcy of companies is an important application of business analytics due to the substantial financial and socio-economic costs incurred. Machine-learning models can help to improve bankruptcy predictions compared to traditional statistical prediction models and thus lower these costs, leading to a better financial outcome for the stakeholders. In this study, we extend the analysis of Barboza et al. (2017) by introducing cost reduction and unexpected job loss as an alternative performance measure for bankruptcy prediction models. This makes it possible to quantify the financial impact the models could have. The 28 companies that filed for bankruptcy in our test dataset (2015 to 2020) caused over 150,000 lost jobs and costs of approximately USD 72,915 million. The most cost-efficient model applied on the data with a three-year delay can reduce the costs by more than 50% with a prediction three years in advance, compared to a baseline model where every company receives a loan.

Our study provides two major contributions. First, we add to the bankruptcy prediction literature by showing that ML models are significantly better than traditional statistical prediction models in terms of both statistical and financial performance. This is true for all sets of input variables and all time delays that we considered in our study. Second, our study indicates that small differences in statistical performance can translate into a substantial difference in financial and socio-economic performance. This highlights the need to include evaluation metrics that properly measure the business impact of the predictions.

We believe that our general idea could be translated to other fields. For multiple binary classification problems, the costs of Type I and Type II errors are different. For instance, in the medical field a false negative prediction often has more severe consequences than a false positive. For some diagnoses, such as certain types of cancer (Houssein et al., 2022) or neurological disorders (Abdel Hameed et al., 2022), an early treatment leads to a significant improvement in patients' outcomes. For example, an early detection of a COVID-19 infection (Bekhet et al., 2020) can prevent the spreading of the virus. Besides the usual statistical measures, algorithms which are used in medical diagnostics could be evaluated based on reduced health spending and increased quality-adjusted life years.

While our study clearly shows the relevance of including financial metrics in the selection of ML models for bankruptcy prediction, we intend to extend this study in several ways. First, particularly as bankruptcy events of large companies can cause shockwaves of different effects across entire industries or economies, we want to additionally consider socio-economic measures of bankruptcy in the analysis of bankruptcy prediction models. ML models with superior prediction capabilities could be beneficial here as a better identification of potential system-relevant bankruptcy events could lower the overall socio-economic impact of potential bankruptcies. Second, we plan to extend the current ML prediction models by including a wider variety of data, e.g., through the integration of textual data. News or company reports could provide relevant information on future bankruptcy events by describing relevant economic trends. Hence, identifying what textual information might be relevant for bankruptcy prediction and augmenting the prediction models with this information could further improve the prediction capabilities of the models. Third, we will also look at additional evaluation measures that aim to quantify the actual socio-economic impact of misclassifications, e.g., in terms of withheld loans, lost capital and workplaces, etc. Finally, we plan to further investigate the temporal aspect of prediction models for bankruptcies. Specifically, we aim to analyze how the prediction performance deteriorates for predictions further into the future, and how far into the future we can reasonably identify bankruptcies. Our current study uses a 3-year forecast window, yet it is unclear if good predictions can also be generated further into the future.

7. Summary conclusion

7.1. Theoretical implications

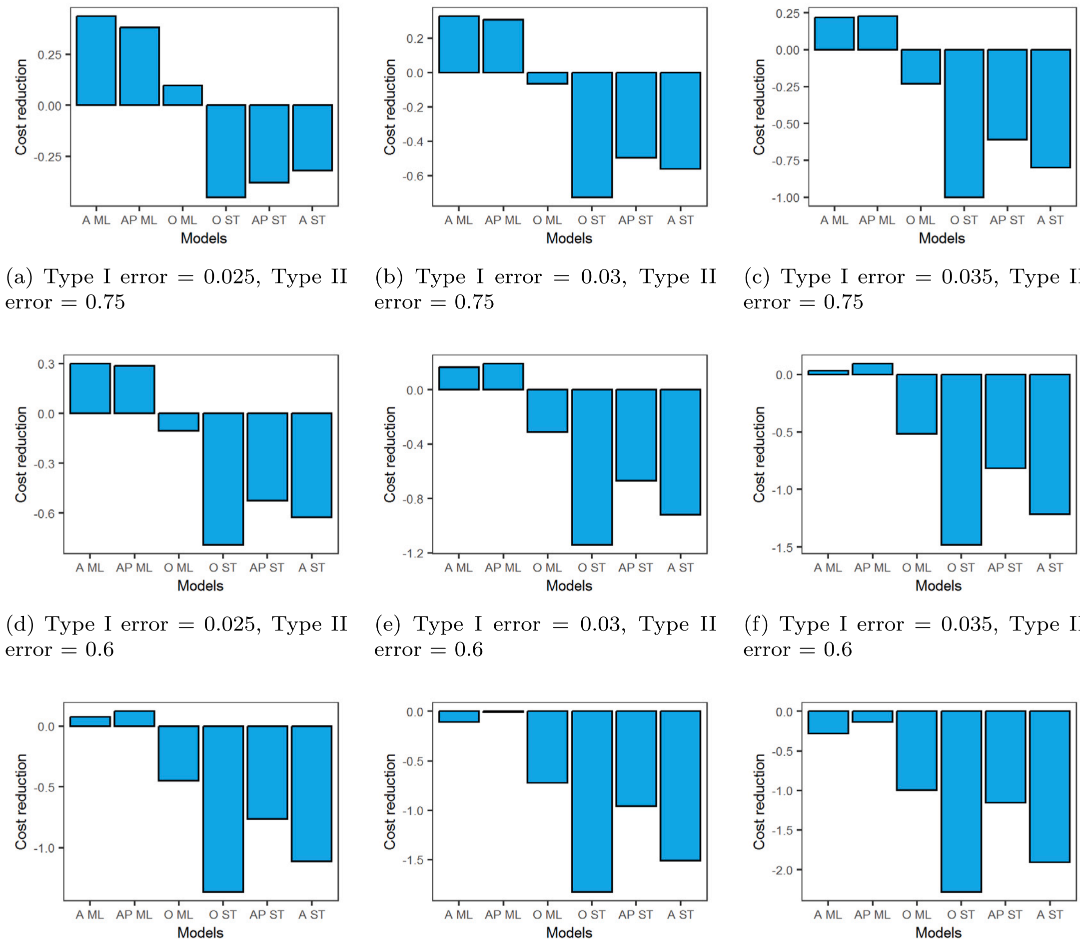
Existing bankruptcy prediction models suffer from the limitation that they are evaluated based only on statistical performance measures, which can be applied to any binary classification problem. However, as our results show, even small differences in these statistical measures can mean large differences in the cost reduction of the models. Because of the different socio-economic costs of Type I and Type II errors, we included them in the model selection. We decided to build upon the research of Barboza et al. (2017) as their study compares a variety of different machine-learning models and uses an extensive dataset about the overall North American corporate credit market. We add to bankruptcy prediction literature by introducing a new approach for evaluating and optimizing bankruptcy prediction models. Practitioners as well as researchers can benefit from our results. Our new metrics help to create and evaluate models in a new, cost-sensitive way, which could lead decision-makers, such as investors and creditors, to make optimal decisions based on their target variable and maximize their profits.

One surprising result is that the Altman and Altman plus variables show similar performance regarding the cost reduction and unpredicted job loss. This could suggest that the additional growth and market variables have a limited predictive power. The five original Altman variables serve as important predictors of financial distress even 50 years after their publication. The Ohlson variables, although they were selected based on a larger sample and a later period, lead to worse financial outcomes and larger unpredicted job loss in almost every alternative scenario.

7.2. Practical implications

Our results are important for several stakeholders. First, companies using bankruptcy prediction models need to take these financial and other aspects into account when building and applying prediction models to minimize the misclassification costs and maximize the saved costs. Second, when the number of employees is used as an additional evaluation criterion, ML models could provide a specific focus on large companies whose bankruptcies could have serious (socio-economic) consequences due to company size, number of employees, or related measures of company relevance from an economic perspective.

In their study about bankruptcy prediction with multi-object classifiers, Zelenkov and Volodarskiy (2021) state that bankruptcy prediction models should have a balanced ratio of Type I and Type II errors, as "there are no reasonable foundations to set different costs for two types of errors in bankruptcy prediction models" (p. 5). They argue that, from a business point of view, losses occur if a contract is signed with a company, which will file for bankruptcy. On the other hand, when a reliable partner is classified as bankrupt, losses might occur because the contract is shifted to a less efficient company. "The exact value of the losses depends on the particular case and it is not always



(a) Type I error = 0.025, Type II error = 0.75
(b) Type I error = 0.03, Type II error = 0.75
(c) Type I error = 0.035, Type II error = 0.75
(d) Type I error = 0.025, Type II error = 0.6
(e) Type I error = 0.03, Type II error = 0.6
(f) Type I error = 0.035, Type II error = 0.6
(g) Type I error = 0.025, Type II error = 0.45
(h) Type I error = 0.03, Type II error = 0.45
(i) Type I error = 0.035, Type II error = 0.45

Fig. A.1. The mean of the different models' AUC, BACC, CR and UJL on all three sets of input variables with a 2-year delay (ML — all machine-learning models, STAT — logistic regression and discriminant analysis, AP — Altman pus variables, A — Altman variables, O — Ohlson variables).

possible to state that the cost of the Type II errors is higher than the cost of the Type I errors" (Zelenkov & Volodarskiy, 2021, p. 5). While this might be true from a company's perspective, we argue that from the perspective of the investor or society, a Type I error has more severe consequences than a Type II error. If investors lends money to a company that goes bankrupt, they can lose the whole investment. Contrarily, if they do not lend money to a solvent company, they lose only the foregone interest, which depending on the demand and supply of loanable funds varies from 2% to 4% of the loan (Altman et al., 1977). From the societal point of view, an unpredicted bankruptcy also has more severe consequences. While a Type II error affects mostly the company that does not receive a loan, a Type I error can have much more severe negative effects. Because of the spreading effect (Jackson & Wood, 2013), one corporate bankruptcy can easily cause the bankruptcy (or severe financial distress) of their customers, competitors or suppliers. After the failure of Lehman Brothers, cross-border lending decreased 58% on average (De Haas & Van Horen, 2012). "Based on Fama–French–Carhart four-factor model adjusted abnormal returns, the 184 equity underwriting clients that we study lost 4.85% of their market value, on average, over a 7-day period spanning the five trading days before and the first and second trading days immediately after Lehman's bankruptcy filing, amounting to approximately USD 23 billion in aggregate risk-adjusted losses" (Fernando et al., 2012, p. 237).

CRediT authorship contribution statement

Jelena Radovanovic: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Christian Haas:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

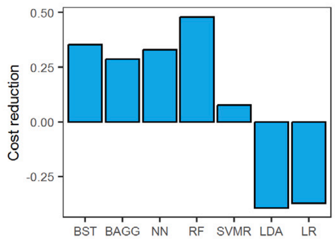
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

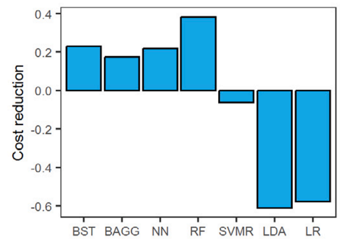
Data will be made available on request.

Appendix. Additional robustness checks with a two year delay

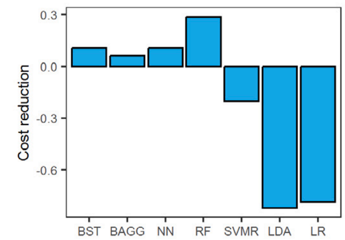
See Figs. A.1 and A.2.



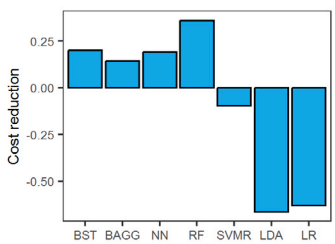
(a) Type I error = 0.025, Type II error = 0.75



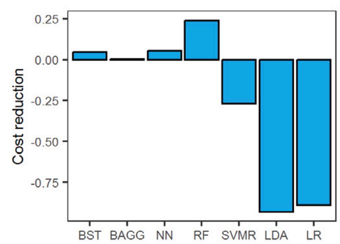
(b) Type I error = 0.03, Type II error = 0.75



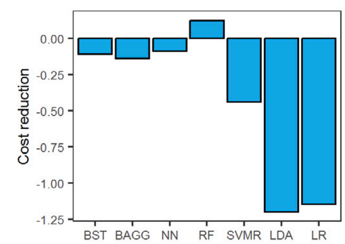
(c) Type I error = 0.035, Type II error = 0.75



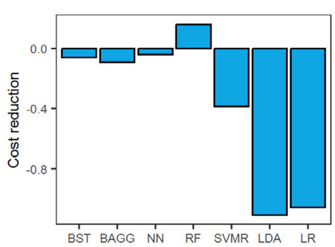
(d) Type I error = 0.025, Type II error = 0.6



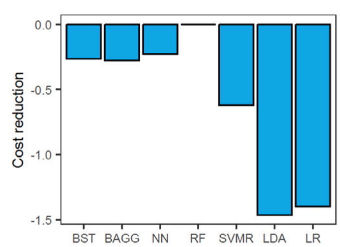
(e) Type I error = 0.03, Type II error = 0.6



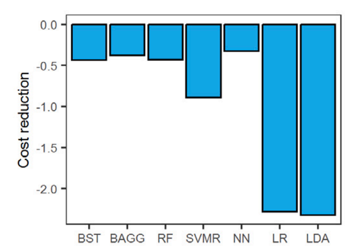
(f) Type I error = 0.035, Type II error = 0.6



(g) Type I error = 0.025, Type II error = 0.45



(h) Type I error = 0.03, Type II error = 0.45



(i) Type I error = 0.035, Type II error = 0.45

Fig. A.2. The mean of the different models' AUC, BACC, CR and UJL on all three sets of input variables with a 2-year delay (BST — Boosting, BAGG — Bagging, RF — random forest, NN — neural network, LR — logistic regression, LDA — linear discriminant analysis).

References

Abdel Hameed, M., Hassaballah, M., Hosney, M. E., & Alqahtani, A. (2022). An AI-enabled internet of things based autism care system for improving cognitive ability of children with autism spectrum disorders. *Computational Intelligence and Neuroscience*, *2022*.

Agarwal, V., & Taffler, R. (2008). Comparing the performance of market-based and accounting-based bankruptcy prediction models. *Journal of Banking & Finance*, *32*(8), 1541–1551.

Al Nasseri, A., Tucker, A., & De Cesare, S. (2015). Quantifying StockTwits semantic terms' trading behavior in financial markets: An effective application of decision tree algorithms. *Expert Systems with Applications*, *42*(23), 9192–9210.

Alaka, H. A., Oyedele, L. O., Owolabi, H. A., Kumar, V., Ajayi, S. O., Akinade, O. O., & Bilal, M. (2018). Systematic review of bankruptcy prediction models: Towards a framework for tool selection. *Expert Systems with Applications*, *94*, 164–184.

Alfaro, E., García, N., Gámez, M., & Elizondo, D. (2008). Bankruptcy forecasting: An empirical comparison of AdaBoost and neural networks. *Decision Support Systems*, *45*(1), 110–122.

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, *23*(4), 589–609.

Altman, E. I., Haldeman, R. G., & Narayanan, P. (1977). ZETATM analysis a new model to identify bankruptcy risk of corporations. *Journal of Banking & Finance*, *1*(1), 29–54.

Altman, E. I., Iwanicz-Drozdowska, M., Laitinen, E. K., & Suvas, A. (2020). A race for long horizon bankruptcy prediction. *Applied Economics*, *52*(37), 4092–4111.

Balcaen, S., & Ooghe, H. (2006). 35 Years of studies on business failure: An overview of the classic statistical methodologies and their related problems. *The British Accounting Review*, *38*(1), 63–93.

Barboza, F., Kimura, H., & Altman, E. (2017). Machine learning models and bankruptcy prediction. *Expert Systems with Applications*, *83*, 405–417.

Beaver, W. H. (1966). Financial ratios as predictors of failure. *Journal of Accounting Research*, 71–111.

Bekhet, S., Hassaballah, M., Kenk, M. A., & Hameed, M. A. (2020). An artificial intelligence based technique for COVID-19 diagnosis from chest X-ray. In *2020 2nd novel intelligent and leading emerging sciences conference* (pp. 191–195). IEEE.

Blöchlinger, A., & Leippold, M. (2006). Economic benefit of powerful credit scoring. *Journal of Banking & Finance*, *30*(3), 851–873.

Boone, A. L., & Ivanov, V. I. (2012). Bankruptcy spillover effects on strategic alliance partners. *Journal of Financial Economics*, *103*(3), 551–569.

Branco, P., Torgo, L., & Ribeiro, R. P. (2016). A survey of predictive modeling on imbalanced domains. *ACM Computing Surveys*, *49*(2), 1–50.

Breiman, L. (1996). Bagging predictors. *Machine Learning*, *24*, 123–140.

Breiman, L. (2001). Random forests. *Machine Learning*, *45*, 5–32.

Carton, R. B., & Hofer, C. W. (2006). *Measuring organizational performance: metrics for entrepreneurship and strategic management research*. Edward Elgar Publishing.

Chen, T.-l., & Chen, F.-y. (2016). An intelligent pattern recognition model for supporting investment decisions in stock market. *Information Sciences*, *346*, 261–274.

Chen, X., Wang, X., & Wu, D. D. (2010). Credit risk measurement and early warning of SMEs: An empirical study of listed SMEs in China. *Decision Support Systems*, *49*(3), 301–310.

Compustat (2022). Compustat from standard & poor's. In *Wharton research data services (WRDS)*, Centennial, CO: Standard & Poor's.

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, *20*, 273–297.

de Basilea, C. d. S. B. (2006). *Basel II: International convergence of capital measurement and capital standards: A revised framework–comprehensive version*. Banco de Pagos Internacionales, junio.

De Haas, R., & Van Horen, N. (2012). International shock transmission after the Lehman Brothers collapse: Evidence from syndicated lending. *American Economic Review*, *102*(3), 231–237.

du Jardin, P. (2016). A two-stage classification technique for bankruptcy prediction. *European Journal of Operational Research*, *254*(1), 236–252.

Fernando, C. S., May, A. D., & Megginson, W. L. (2012). The value of investment banking relationships: Evidence from the collapse of Lehman Brothers. *The Journal of Finance*, *67*(1), 235–270.

Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, *7*(2), 179–188.

Fitzpatrick, P. J. (1932). *A comparison of the ratios of successful industrial enterprises with those of failed companies*.

Freund, Y. (1995). Boosting a weak learning algorithm by majority. *Information and Computation*, *121*(2), 256–285.

Friedman, J. H. (2002). Stochastic gradient boosting. *Computational Statistics & Data Analysis*, *38*(4), 367–378.

Geng, R., Bose, I., & Chen, X. (2015). Prediction of financial distress: An empirical study of listed Chinese companies using data mining. *European Journal of Operational Research*, *241*(1), 236–247.

Gentry, J. A., Newbold, P., & Whitford, D. T. (1985). Classifying bankrupt firms with funds flow components. *Journal of Accounting Research*, 146–160.

Ghatasheh, N., Faris, H., Abukhurma, R., Castillo, P. A., Al-Madi, N., Mora, A. M., Al-Zoubi, A. M., & Hassanat, A. (2020). Cost-sensitive ensemble methods for bankruptcy prediction in a highly imbalanced data distribution: A real case from the Spanish market. *Progress in Artificial Intelligence*, *9*, 361–375.

Giordani, P., Jacobson, T., Von Schedvin, E., & Villani, M. (2014). Taking the twists into account: Predicting firm bankruptcy risk with splines of financial ratios. *Journal of Financial and Quantitative Analysis*, *49*(4), 1071–1099.

Grice, J. S., & Dugan, M. T. (2001). The limitations of bankruptcy prediction models: Some cautions for the researcher. *Review of Quantitative Finance and Accounting*, *17*(2), 151–166.

Hillegeist, S. A., Keating, E. K., Cram, D. P., & Lundstedt, K. G. (2004). Assessing the probability of bankruptcy. *Review of Accounting Studies*, *9*(1), 5–34.

Houssein, E. H., Abdelkareem, D. A., Emam, M. M., Hameed, M. A., & Younan, M. (2022). An efficient image segmentation method for skin cancer imaging using improved golden jackal optimization algorithm. *Computers in Biology and Medicine*, *149*, Article 106075.

Jackson, R. H., & Wood, A. (2013). The performance of insolvency prediction and credit risk models in the UK: A comparative study. *The British Accounting Review*, *45*(3), 183–202.

Kolay, M., Lemmon, M., & Tashjian, E. (2016). Spreading the misery? Sources of bankruptcy spillover in the supply chain. *Journal of Financial and Quantitative Analysis*, *51*(6), 1955–1990.

Krauss, C., Do, X. A., & Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, *259*(2), 689–702.

Lee, S., & Choi, W. S. (2013). A multi-industry bankruptcy prediction model using back-propagation neural network and multivariate discriminant analysis. *Expert Systems with Applications*, *40*(8), 2941–2946.

Li, S., Wang, M., & He, J. (2013). Prediction of banking systemic risk based on support vector machine. *Mathematical Problems in Engineering*, *2013*.

Messier, W. F., & Hansen, J. V. (1988). Inducing rules for expert system development: An example using default and bankruptcy data. *Management Science*, *34*(12), 1403–1415.

Neves, J. C., & Vieira, A. (2006). Improving bankruptcy prediction with hidden layer learning vector quantization. *European Accounting Review*, *15*(2), 253–271.

Noble, W. S. (2006). What is a support vector machine? *Nature biotechnology*, *24*(12), 1565–1567.

Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, 109–131.

Olson, D. L., Delen, D., & Meng, Y. (2012). Comparative analysis of data mining methods for bankruptcy prediction. *Decision Support Systems*, *52*(2), 464–473.

Ooghe, H., & De Prijcker, S. (2008). *Failure processes and causes of company bankruptcy: a typology*. Management Decision, Emerald Group Publishing Limited.

Qian, H., Wang, B., Yuan, M., Gao, S., & Song, Y. (2022). Financial distress prediction using a corrected feature selection measure and gradient boosted decision tree. *Expert Systems with Applications*, *190*, Article 116202.

Shi, Y., & Li, X. (2019). An overview of bankruptcy prediction models for corporate firms: A systematic literature review. *Intangible Capital*, *15*(2), 114–127.

Shin, K.-S., Lee, T. S., & Kim, H.-j. (2005). An application of support vector machines in bankruptcy prediction model. *Expert Systems with Applications*, *28*(1), 127–135.

Tobback, E., Bellotti, T., Moeyersoms, J., Stankova, M., & Martens, D. (2017). Bankruptcy prediction for SMEs using relational data. *Decision Support Systems*, *102*, 69–81.

Veganzones, D., & Séverin, E. (2018). An investigation of bankruptcy prediction in imbalanced datasets. *Decision Support Systems*, *112*, 111–124.

World Bank (2022). Risk premium on lending (lending rate minus treasury bill rate). URL: https://data.worldbank.org/indicator/FR.INR.RISK.

Yeh, C.-C., Chi, D.-J., & Lin, Y.-R. (2014). Going-concern prediction using hybrid random forests and rough set approach. *Information Sciences*, *254*, 98–110.

Zelenkov, Y., & Volodarskiy, N. (2021). Bankruptcy prediction on the base of the unbalanced data using multi-objective selection of classifiers. *Expert Systems with Applications*, *185*, Article 115559.

Zhao, Z., Xu, S., Kang, B. H., Kabir, M. M. J., Liu, Y., & Wasinger, R. (2015). Investigation and improvement of multi-layer perceptron neural networks for credit scoring. *Expert Systems with Applications*, *42*(7), 3508–3516.

Zmijewski, M. E. (1984). Methodological issues related to the estimation of financial distress prediction models. *Journal of Accounting Research*, 59–82.